FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 4.1: GREENHOUSE EFFECT AND CLIMATE CHANGE	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

WHAT IS THIS DOCUMENT?

This is your Final Explanation guide. Over the past 7 lessons, you have investigated why Mount Kenya's Lewis Glacier is shrinking and why farmers in Embu County can no longer predict the long rains. Now it is time to bring all of your evidence, models, and reasoning together into one powerful, well-structured scientific explanation.

YOUR TASK:

Write a full scientific explanation that answers the driving question:

"Why are Mount Kenya's glaciers melting and Kenyan farmers' rains becoming unpredictable — and what can the physics of our atmosphere tell us about how to respond?"

HOW TO USE THIS DOCUMENT:

- Read each section prompt carefully before you write.
- Use the exemplar response in each section as a model — notice the scientific vocabulary, the use of evidence, and the way ideas are linked together.
- Your explanation must connect all the way from the physics of infrared radiation to the real-world changes happening on Mount Kenya and on farms across Kenya.
- Write in full sentences. Use the key terms from your lessons. Reference specific evidence you gathered — from the Lewis Glacier images, the Embu farmer testimony, your greenhouse gas simulations, and your energy-balance investigations.
- You are not copying the exemplar — you are using it as a standard to aim for and then writing your own original explanation.

KEY TERMS YOU SHOULD USE:

electromagnetic radiation, infrared radiation, greenhouse gases, carbon dioxide (CO_2), methane (CH_4), water vapour, absorption, re-emission, energy balance, global mean surface temperature, glacier retreat, albedo, precipitation patterns, feedback loop, mitigation, adaptation

KICD LEARNING OUTCOMES THIS ADDRESSES:

By completing this Final Explanation, you demonstrate that you can:

1. Explain how the greenhouse effect influences global and local climates.
2. Analyse the causes and effects of climate change and propose measures to address it.

Section 1: The Physics of the Greenhouse Effect	
Explain, using the physics of electromagnetic radiation and molecular absorption, how the greenhouse effect works. Your explanation should answer: What kinds of radiation does the Sun send to Earth? What happens to that energy at Earth's surface? How do greenhouse gas molecules interact with infrared radiation, and why does this warm the lower atmosphere? Use a step-by-step energy-flow argument.	The Sun emits energy across a wide spectrum of electromagnetic radiation, but most of the energy that reaches Earth's surface arrives as visible light and ultraviolet radiation, which have short wavelengths and high energy. When this radiation strikes Earth's surface — the volcanic rock and ice of Mount Kenya, the red soil of a maize farm in Embu, or the surface of Lake Victoria — the surface absorbs it and warms up. A warm surface then re-emits energy, but it re-emits it as infrared radiation, which has much longer wavelengths and lower energy than the incoming visible light. This is where the physics of greenhouse gases becomes critical. Molecules of carbon dioxide (CO_2), methane (CH_4), and water vapour have a special property: their molecular bonds can absorb infrared radiation and vibrate. When they absorb this outgoing infrared energy, they re-emit it in all directions — including back downward toward Earth's surface. This process is called the greenhouse effect, and it means that energy that would otherwise escape to space is instead trapped in the lower atmosphere, warming the surface. Without any greenhouse effect, Earth's average surface temperature would be approximately –18 °C — far too cold to support the maize, beans, and tea that Kenyan farmers grow. The natural greenhouse effect is therefore essential to life. The problem we are investigating is what happens when the concentration of these greenhouse gases increases beyond natural levels.

Section 2: Human Activities, Rising CO_2, and the Disruption of Energy Balance	
Explain how human activities have increased the concentration of greenhouse gases in the atmosphere, and describe what effect this has on Earth's energy balance. Use the concept of radiative forcing and refer to specific sources of greenhouse gas emissions that are relevant in the Kenyan and global context. Why does a small change in greenhouse gas concentration have such a significant effect on temperature?	Earth's climate is stable when the amount of energy arriving from the Sun equals the amount of energy escaping back to space — this is called energy balance. Human activities since the industrial era have released enormous quantities of greenhouse gases that did not previously exist in the atmosphere at such high concentrations. The burning of fossil fuels (petrol, diesel, coal) releases CO_2; large-scale deforestation — including the clearing of forests in the Mau Complex in Kenya's Rift Valley — removes trees that would otherwise absorb CO_2 through photosynthesis; agricultural expansion and livestock farming release methane. Globally, atmospheric CO_2 levels have risen from approximately 280 parts per million (ppm) before industrialisation to over 420 ppm today. This increase disrupts Earth's energy balance. With more greenhouse gas molecules in the atmosphere, more outgoing infrared radiation is absorbed and re-emitted back toward the surface before it can escape to space. This is called an increase in radiative forcing — more energy is being retained by the Earth system than is leaving it. Even a seemingly small increase in the global mean surface temperature of 1–1.5 °C represents an enormous amount of extra thermal energy distributed across the entire planet. This extra energy does not affect all places equally — it accelerates ice melt at high altitudes like Mount Kenya, shifts atmospheric pressure patterns, and changes the timing and intensity of rainfall systems that Kenyan farmers depend on.

Section 3: Connecting the Physics to Mount Kenya's Glaciers and Kenya's Rainfall	
Now apply the physics you have explained to the two specific changes in the anchoring phenomenon: (a) the retreat of the Lewis Glacier on	The retreat of Mount Kenya's Lewis Glacier is a direct physical consequence of rising atmospheric temperatures caused by the enhanced greenhouse effect. Glacial ice is maintained when the rate of snowfall and freezing equals or exceeds the rate of melting. As the lower atmosphere warms, the freezing level — the altitude at which temperatures drop below 0 °C — rises higher on the mountain. This means less of the mountain remains cold enough to sustain glacier ice year-round. The 1934 photograph of the Lewis Glacier and the current satellite image show dramatic evidence of this: the glacier has lost the majority of its surface

Mount Kenya, and (b) the increasingly unpredictable long rains experienced by the maize farmer in Embu County. Explain the physical mechanisms behind each change, and show how both are caused by the same underlying process. You may refer to feedback loops where relevant.	area, and scientists estimate it could disappear entirely within decades. Glacier retreat also creates a feedback loop that accelerates warming. Ice has a high albedo — it reflects a large fraction of incoming solar radiation back to space. When ice is replaced by dark volcanic rock, the exposed rock absorbs much more solar radiation, warming the local environment further and causing even faster melting. The disruption of the long rains in Embu County is connected to the same root cause through atmospheric physics. Kenya's rainfall patterns, including the March–May long rains, are driven by the movement of the Inter-Tropical Convergence Zone (ITCZ) — a band of low pressure and heavy rainfall that moves northward and southward across the equator with the seasons. Rising global temperatures alter atmospheric pressure gradients and sea-surface temperatures in the Indian Ocean. These changes shift the position and timing of the ITCZ, causing the long rains to arrive later, become shorter, or fail entirely in some years. The Embu farmer's testimony — that rains now arrive weeks later than in their grandparents' time — is consistent with the scientific data on shifting rainfall patterns across East Africa. Both the glacial retreat and the rainfall disruption are therefore symptoms of the same cause: an enhanced greenhouse effect driven by rising greenhouse gas concentrations.

Section 4: Evidence That Supports This Explanation	
A strong scientific explanation is built on evidence, not just theory. Identify and describe at least THREE pieces of evidence — from your lessons, investigations, or the anchoring phenomenon — that support the explanation you have built. For each piece of evidence, explain what it shows and how it connects to the physics of the greenhouse effect and climate change.	First, the paired images of Mount Kenya's Lewis Glacier — the 1934 photograph and the current satellite image — provide direct visual evidence of glacier retreat over time. This is consistent with the physics prediction that rising temperatures caused by the enhanced greenhouse effect would cause high-altitude glaciers to melt. The rate of retreat has accelerated in recent decades, which aligns with the period of most rapid increase in atmospheric CO_2. Second, in our energy-balance simulation investigation, we modelled what happens to the temperature of a system when the concentration of an infrared-absorbing gas is increased. The simulation showed that increasing CO_2 concentration caused a measurable rise in the equilibrium temperature of the model atmosphere. This is a controlled, repeatable demonstration of the physical mechanism — not just a correlation observed in nature. Third, the testimony of the maize farmer in Embu County provides real-world observational evidence of a changing climate at the local level. The farmer reports that the long rains now arrive weeks later than in previous generations and that maize and bean yields have fallen. This is consistent with scientific records from Kenya Meteorological Department data showing that East African rainfall patterns have become less predictable since the 1970s — a period that closely matches the acceleration of global greenhouse gas emissions. Fourth, the molecular absorption model — which shows that CO_2 and CH_4 molecules absorb infrared radiation at specific wavelengths — explains the physical mechanism at the atomic scale. This connects the large-scale observations (glacier retreat, rainfall shifts) to a fundamental, well-understood principle of physics: the interaction between electromagnetic radiation and molecular bonds.

Section 5: Physics-Informed Responses — Mitigation and Adaptation	
Understanding the physics of the greenhouse effect and	Mitigation strategies aim to reduce the amount of greenhouse gases being added to the atmosphere, addressing the root physical cause of climate change.

climate change is not just about explaining what is happening — it also guides how we respond. Describe TWO mitigation strategies and TWO adaptation strategies that are relevant to Kenya, and explain how each one is grounded in the physics you have studied. Then reflect: what is the most important thing that the physics of the atmosphere tells us about the urgency of action?	One important mitigation strategy in Kenya is protecting and restoring forests, such as the rehabilitation of the Mau Forest Complex in the Rift Valley. Trees absorb CO_2 from the atmosphere through photosynthesis, converting it into stored carbon in wood and soil. By removing CO_2 from the atmosphere, reforestation reduces the concentration of greenhouse gases and partially restores the energy balance. This directly addresses the physical mechanism that is causing warming. A second mitigation strategy is the transition to renewable energy sources — such as Kenya's world-leading geothermal energy from Olkaria in the Rift Valley, and expanding solar and wind energy. Burning fossil fuels releases CO_2 that was stored underground for millions of years, rapidly increasing atmospheric concentration. Replacing fossil fuels with renewable sources stops this additional CO_2 from entering the atmosphere, reducing radiative forcing. Adaptation strategies do not reduce warming, but help communities adjust to the changes that are already occurring. One adaptation strategy for Kenyan farmers is the adoption of drought-resistant maize varieties and diversified crops such as sorghum and millet, which are more tolerant of late or reduced rainfall. This does not change the physics of the atmosphere, but it acknowledges that rainfall patterns have already shifted and builds resilience into food systems. A second adaptation strategy is the construction of water harvesting systems — such as terraced hillside farms and rainwater storage tanks — that capture rainfall when it does arrive and store it for use during dry spells. This is especially important given the physics finding that rainfall is becoming more intense but less frequent in parts of Kenya. The most important insight from the physics is this: greenhouse gases such as CO_2 remain in the atmosphere for hundreds of years after they are emitted. This means that the warming we observe today is partly the result of emissions from decades ago, and the emissions we produce today will continue to affect the climate long into the future. The physics of feedback loops — where warming causes further warming, as in the glacier-albedo feedback — also tells us that delay makes the problem significantly harder to reverse. Understanding this physics is not just an academic exercise: it is essential knowledge for every Kenyan student who will live and make decisions in a changing climate.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Greenhouse Effect Explanation	Accurately and thoroughly explains the full energy-flow pathway: incoming solar radiation → surface absorption → infrared re-emission → greenhouse gas molecular absorption and re-emission → surface warming. Correct use of all key terms (infrared radiation, absorption, re-emission, radiative forcing, energy balance). No scientific errors.	Explains the main steps of the greenhouse effect with mostly correct scientific language. May omit one step (e.g., the distinction between incoming and outgoing radiation wavelengths) or use one key term imprecisely. No major misconceptions.	Shows a basic understanding that greenhouse gases trap heat, but the explanation is incomplete, vague, or contains a scientific error (e.g., confusing the greenhouse effect with ozone depletion, or not explaining why the surface emits infrared radiation).
Connection of Physics to the	Explicitly and clearly connects the	Connects the physics to both parts of the	Connects the physics to only one part of

Anchoring Phenomenon	physics of the enhanced greenhouse effect to BOTH the retreat of the Lewis Glacier AND the disruption of the long rains in Embu County. Explains the physical mechanism behind each change (e.g., rising freezing level for glacier retreat; ITCZ shift for rainfall disruption). Identifies the shared root cause.	anchoring phenomenon, but the physical mechanism for one of the two changes is explained less clearly or with less precision. The shared root cause is identified.	the anchoring phenomenon (either glacier retreat or rainfall disruption), or describes the connection in general terms without explaining the specific physical mechanism.
Quality and Use of Evidence	Identifies and explains at least THREE distinct pieces of evidence (observational, experimental, and/or data-based). For each, clearly states what the evidence shows and explicitly links it to the physics explanation. Evidence is specific (e.g., references the 1934 vs. current Lewis Glacier images, the energy-balance simulation, the farmer testimony, or Kenya Meteorological data).	Identifies at least TWO pieces of evidence and links them to the physics explanation. Evidence is mostly specific but may lack a clear statement of how one piece of evidence supports the physics mechanism.	Mentions evidence but in a vague or general way (e.g., 'we saw that temperatures are rising' without specifying which evidence or what it shows). Fewer than two distinct pieces of evidence are used, or evidence is not clearly connected to the physics.
Mitigation and Adaptation Responses Grounded in Physics	Describes TWO mitigation and TWO adaptation strategies relevant to Kenya. For each strategy, clearly explains how it is grounded in the physics studied (e.g., reforestation reduces atmospheric CO_2 concentration → reduces radiative forcing; renewable energy stops additional CO_2 emissions → restores energy balance). Distinguishes correctly between mitigation (addressing root cause) and adaptation (adjusting to effects).	Describes at least one mitigation and one adaptation strategy with a correct Kenya-relevant example. The physics connection is explained for most strategies but may be incomplete for one. The distinction between mitigation and adaptation is mostly clear.	Describes responses to climate change but does not clearly distinguish between mitigation and adaptation, or does not connect the strategies to the physics studied. Examples may be generic and not specific to Kenya.
Scientific Communication and Use of Key Terminology	Explanation is clearly structured, logically sequenced, and written in full sentences throughout. Consistently uses key scientific terms correctly and in context (electromagnetic radiation, infrared radiation, greenhouse gases, albedo, feedback loop, energy balance, radiative forcing, etc.). Arguments flow from evidence to reasoning to conclusion in a way that is easy to follow.	Explanation is mostly clear and logically organised. Most key terms are used correctly, though one or two may be used imprecisely or inconsistently. The argument is generally easy to follow with minor gaps in logical flow.	Explanation is incomplete, disorganised, or difficult to follow in places. Few key scientific terms are used, or they are used incorrectly. The response reads more as a list of facts than a connected scientific argument.